Rohan Tandel

Senior Backend Engineer (Node.js & Distributed Systems)

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 8+ years of experience architecting resilient distributed systems, asynchronous event-driven pipelines, and high-concurrency cloud backends using Node.js, TypeScript, and AWS. Architected transactional outbox auto-submission engines with randomized jitter, decoupled ingestion pipelines via AWS SQS with DLQ recovery, engineered OpenID Connect identity services with RS256 token signing, and tuned high-volume PostgreSQL/Redis data layers. Proven reliability lead driving deterministic state machines and robust multi-tenant architectures.

CORE CAPABILITIES & TECHNICAL EXPERTISE

Backend Architecture
Node.js · Express · TypeScript · Microservices · REST API Design

Distributed Systems & Queues
AWS SQS · DLQ Recovery · Transactional Outbox · Redis Caching & Pub/Sub

Data Engineering & Storage
PostgreSQL · Query Optimization · Index Tuning · MongoDB · Data Modeling

Security & Identity
OpenID Connect (OIDC) · OAuth2 · JWKS · RS256 Signing · RBAC · Multi-Tenancy

Cloud Infrastructure
AWS (ECS / Fargate, Lambda, SNS, API Gateway, CloudFront) · Docker · CI/CD

AI Integration & Workflows
LLM Orchestration · Prompt Design · Structured Outputs · Tool Calling · Guardrails

PROFESSIONAL EXPERIENCE

Lead Reliability & Engineering · Sr. Backend & Integration Engineer

14 Jul 2022 — 13 Jul 2026

Fintua · European TaxTech & Fintech SaaS (Remote)

Led reliability engineering and served as Scrum Lead. Architected core compliance infrastructure, OpenID Connect identity services, outbox submission engines, and metadata-driven frameworks across European jurisdictions.

- **Configurable Return Data & Table Framework:** Designed and developed a backend-metadata-driven configuration framework for reusable data tables and return-specific XML data definitions. Standardized dependencies, filters, and field rules across countries and return types, reducing repetitive conditional development and cutting delivery time for new tables and return-data structures by approximately 70%.
- **Shared Transaction Ingestion Across Return Types:** Reworked transaction ingestion to decouple raw transaction storage from individual return lifecycles. Enabled multiple related return types to process shared transaction data while maintaining independent catch rules, audit trails, editing permissions, and status transitions. Integrated SQS, DLQ recovery, Redis caching, and PostgreSQL query optimizations.
- **Automated Return Submission Service:** Engineered an automated tax-return submission service using a scheduler-triggered transactional outbox pattern. Managed tax-office API submissions, response collection, exponential backoff retries with randomized jitter, and structured submission states, while retaining manual submission as a fallback.
- **OpenID Connect Identity Infrastructure & E-Invoice:** Configured and implemented Fintua's OpenID Connect identity provider (OIDC discovery, Authorization Code flow, JWKS, UserInfo, and RS256 token signing). Developed supporting client data-upload APIs and authenticated navigation between Compliance and the E-Invoice product.
- **B2B Subdomain & Partner Onboarding:** Implemented an end-to-end B2B partner onboarding and authentication workflow, supporting both subdomain-based and standard configurations with partner-specific login, email setup, and UI routing.
- **Standalone Expense Management Integration (Ramp):** Independently developed an end-to-end integration with Ramp as a solo initiative, implementing OAuth authentication and API connectivity to ingest company expense transactions into the expense management workflow.
- **Return Lifecycle & 8-Stage Status Architecture:** Restructured the return lifecycle status system into 8 deterministic states (Data provided → Prepare → Reviewed → Approve → Submit → Submitting / Submission failed → Submitted to Tax Authority) with country-specific event transitions.
- **Engineering Leadership & Reliability:** Led the Reliability & Engineering team and served as Scrum Lead. Optimized CI/CD deployment pipelines, conducted low-level design (LLD) reviews, and mentored engineers through feature development and delivery.

Node.js | TypeScript | AWS (Lambda, SQS, ECS) | Transactional Outbox | PostgreSQL | Redis | OpenID Connect | Docker

Senior Software Engineer (Backend Systems)

02 Jul 2020 — 31 May 2022

Tech Mahindra Pvt Ltd

Engineered distributed synchronization systems for offline educational software and developed broadband customer onboarding workflows for British Telecom.

- **OfflineSync (Educational Initiative / Mindspark):** Developed a bidirectional data synchronization system for offline Mindspark deployments in field schools with limited bandwidth. Engineered schedulers, retry logic, versioning, JSON data compression, remote updates/rollbacks, and an operational dashboard for remote monitoring and manual sync triggers.
- **Plusnet Broadband ISP (British Telecom):** Developed customer-facing dashboard features for broadband connection requests, address validation support systems, and mini approval workflow engines.
- **Mindspark Core Maintenance:** Delivered core learning feature enhancements, production bug resolutions, and legacy codebase migrations.

Node.js | PHP / Laravel | MySQL | Offline Sync | REST APIs | Distributed Data

Software Developer (Backend & Integrations)

04 Jun 2018 — 02 Jul 2020

Scriptbees Pvt Ltd

Built authentication, academic credential systems, and enterprise Salesforce integrations.

- **Client Greenlight:** Engineered JWT authentication systems, academic transcript management workflows, and multi-tenant organization onboarding.
- **VolunteerMarks:** Integrated Salesforce APIs, implemented user check-in and check-out tracking mechanisms, and developed automated email and notification delivery engines.

Node.js | Express | MongoDB | Salesforce API | JWT | REST APIs

INDEPENDENT PLATFORMS & ARCHITECTURE INITIATIVES

Nexora — Configurable Data Platform SaaS ↗

Platform Engineering · Active Development

Live Platform: rohan-tandel.github.io/nexora-saas

A metadata-driven data platform that standardizes dynamic table rendering, column behaviors, filtering, sorting, permissions, and entity schemas across business applications—eliminating repetitive conditional development.

TypeScript | Angular | Node.js | PostgreSQL | Tailwind CSS | API Design

Relay (formerly Kooey Recover) — Recovery Operations & Resolution Platform ↗

Product & Platform Architecture · Active Development

Live Platform: rohan-tandel.github.io/kooey-recover

A capability-driven, multi-tenant recovery operations platform connecting financial institutions and recovery agencies. Focused on dispute tracking, batch workflows, automated scheduling rules, and payment resolution lifecycles.

Node.js | TypeScript | PostgreSQL | Multi-tenancy | Redis | Modular Monolith